

# On the pathwidth of chordal graphs

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## *Abstract*

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In this paper we first show that the pathwidth problem for chordal graphs is NP-hard.

Then we give polynomial algorithms for subclasses. One of those classes are the  $k$ -starlike graphs – a generalization of split graphs. The other class are the primitive starlike graphs – a class of graphs where the intersection behavior of maximal cliques is strongly restricted.

## 1. Overview

The pathwidth problem — PWP for short — has been studied in various fields of discrete mathematics. It asks for the size of a minimum path decomposition of a given graph.

There are many other problems which have turned out to be equivalent (or nearly equivalent) formulations of our problem:

- the interval graph extension problem,
- the gate matrix layout problem,
- the node search number problem,
- the edge search number problem,

see, e.g. [13, 14] or [17]. The first three problems are easily seen to be reformulations. For the fourth there is an easy transformation to the third [14]. Section 2 introduces the problem as well as other problems and classes of graphs related to it. Section 3 gives basic facts on path decompositions.

PWP is NP-hard as was first shown by Kashiwabara and Fujisawa [12] using the

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formulation of interval graph extension. Arnborg, Corneil and Proskurowski [1] were the first to show it in the formulation given here.

Monien and Sudborough [18] have shown that edge search is NP-hard even for planar graphs with vertex degree at most three. The transformation to node search given by Kirousis and Papadimitriou in [14] preserves planarity and degree constraints. So PWP is even NP-hard for this class of graphs.

On the other hand little is known about subclasses of graphs for which the problem is polynomial. The only classes known are trees (see [13, 15, 22]) and cographs [3].

Especially the algorithms known for trees are complicated and give little hope for generalizations, e.g., to  $k$ -trees. Also general approaches to treelike structures as, e.g., given by Arnborg, Lagergren and Seese [2] cannot be applied directly to the pathwidth of  $k$ -trees.

Since many problems become tractable when restricted to chordal graphs one would expect that this would hold for PWP, too. In contrast to that we show in Section 4 that PWP is NP-hard for the class of chordal graphs respectively for a special subclass, the starlike graphs. A starlike graph is a chordal graph that has one central clique and whose other maximal (so-called peripheral) cliques intersect only with this central clique.

The reduction given for the proof starts with a graph partitioning problem, the so-called vertex separator problem. In Sections 5–7 we give algorithmic results for subclasses of the class of starlike graphs. The algorithms we give work on restricted versions of such partitioning problems.

One subclass we solve is the class of primitive starlike graphs. It corresponds to the starlike graphs whose peripheral cliques do not intersect. Section 5 shows that the pathwidth of a primitive starlike graph can be calculated in  $O(|V(G)|^2)$  time and space. For the proof we show that this problem is equivalent to a generalized partition problem on natural numbers, and we derive a pseudopolynomial algorithm for it. This algorithm is an extension of the algorithm for the classical partition problem given in [6].

In Section 6, this algorithm is extended to an optimal algorithm for starlike graphs. It has exponential running time, where the exponent depends on the number of certain subsets of the central clique.

This algorithm is then used in Section 7 to show that the pathwidth of a  $k$ -starlike graph can be calculated in  $O(|V(G)|^{2k+1})$  time and space.  $k$ -starlike graphs are starlike graphs where the size of each clique minus the central clique is bounded by a constant  $k$ . Since split graphs are the 1-starlike graphs we see that pathwidth for split graphs can be calculated in  $O(|V(G)|^3)$  time.

## 2. Basic definitions

We are mainly concerned with three different optimization problems on graphs and with special classes of graphs related to these problems. All the definitions we

give deal with decompositions of the vertex set of a graph into subsets such that each edge will be completely contained in one of the subsets. To point out the close relations between all our definitions we try to formulate them in a unified way.

We start with the definition of the pathwidth problem as it was given by Robertson and Seymour in [20].

**Definition 2.1 (PWP).** The *pathwidth problem* — PWP for short — is the following:

*Instance:*  $G = (V, E)$  a (finite) undirected graph.

*Problem:* Find sets  $X_i \subseteq V$ ,  $i \in I = \{1, \dots, r\}$ , such that  $\max_{i \in I} |X_i|$  is minimum with the following properties:

(W1)  $\bigcup_{i \in I} X_i = V$ .

(W2) For every  $\{v, w\} \in E$  there is an  $i \in I$  with  $\{v, w\} \subseteq X_i$ .

(W3) For every  $i, j, k \in I$ ,  $j$  between  $i$  and  $k$ ,  $X_i \cap X_k \subseteq X_j$ .

In (W3) “ $j$  between  $i$  and  $k$ ” means  $i \leq j \leq k$ . Observe that (W1) is redundant if  $G$  has no isolated vertices.

We will call a feasible solution for PWP of  $G$  a *path decomposition* of  $G$  and the value of an optimal solution minus 1 is the *pathwidth* of  $G$ .

One basis of our discussions will be the following remark.

**Remark 2.2** (clique containment lemma). Let  $(X_i)_{i \in I}$  be a path decomposition of graph  $G$ , and let  $C$  be a clique of  $G$ . Then for  $i_c = \min_{v \in C} \max_{v \in X_i} \{i\}$ , we have  $C \subseteq X_{i_c}$ .

The proof is a straightforward application of properties (W2) and (W3).

The following class of graphs is closely related to PWP:

**Definition 2.3.**  $G = (V, E)$  is called an *interval graph* iff there are  $X_i \subseteq V$ ,  $i \in I = \{1, \dots, r\}$  which fulfill (W1), (W3) and (W2’):

(W2’)  $e \in E$  iff there is  $i \in I$  with  $e \subseteq X_i$ .

This definition is easily seen to be equivalent to the classical definition of interval graphs (see [8]). Just take

$$\left[ \min_{v \in X_i} i, \max_{v \in X_i} i \right]$$

as interval for vertex  $v$ .

For a survey of interval graphs see [9, 16].

PWP can be seen as looking for an interval graph extension of  $G$  with minimum clique size. From that point of view, Remark 2.2 expresses the Helly property of intervals.

Robertson and Seymour (in [21]) also considered a relaxation of PWP:

**Definition 2.4 (TWP).** The *treewidth problem*—TWP for short—is the following:

*Instance:*  $G=(V,E)$ .

*Problem:* Find sets  $X_i \subseteq V$ ,  $i \in I$ , and a tree  $T=(I, E(T))$ , such that  $\max_{i \in I} |X_i|$  is minimal subject to (W1)–(W3).

Here the term “between” in (W3) is understood as  $j$  lying on the (unique) path of  $T$  between  $i$  and  $k$ .

We will call a feasible solution for TWP of  $G$  a *tree decomposition* of  $G$ .

A class of graphs related to TWP is given by

**Definition 2.5.** A graph  $G=(V,E)$  is called *chordal* if there are  $X_i \subseteq V$ ,  $i \in I$  and a tree  $T=(I, E(T))$  which fulfill (W1), (W2') and (W3).

TWP can be seen as searching for an extension of  $G$  to a chordal graph with minimum clique size.

Notice that the classical definitions of interval and chordal graphs are quite different but the equivalence to Definitions 2.3 and 2.5 follows easily from [8], respectively [23], [7] and [4].

More about chordal graphs can be found in [9].

The third problem considered in this paper is given by the following definition. We will use it in Section 4 for a reduction proof.

**Definition 2.6 (VSP).** The *vertex separator problem*—VSP for short—is the following:

*Instance:*  $G=(V,E)$ .

*Problem:* Find sets  $X_1, X_2 \subseteq V$ , that fulfill (W1) and (W2) and minimize  $\max\{|X_1|, |X_2|\}$ .

The name of the problem is motivated by the fact that  $X_1 \cap X_2$  separates  $G$ .

In the sequel we will not distinguish between the optimization problems above and the related decision problems asking for the existence of sets with  $\max_{i \in I} |X_i| \leq k$ , for a natural number  $k$  as additional input.

**Definition 2.7.** (1) A chordal graph  $G$  is called *starlike* iff it has a tree decomposition  $(T, \{X_0, \dots, X_r\})$  so that every set induces a maximal clique and the tree  $T$  is a star.

(2) A starlike graph  $G$  is called *primitive starlike* if

(a)  $X_i \cap X_j = \emptyset$ ,  $\forall i, j \neq 0$ ,

(b)  $X_0 \subseteq \bigcup_{i \neq 0} X_i$ .

(3) For a fixed tree decomposition  $(T, \{X_0, \dots, X_r\})$  of a starlike graph  $G$  put  $\alpha_i = |X_i \cap X_0|$  and  $\beta_i = |X_i \setminus X_0|$ .

(4) We call the vertices in  $X_0$  respectively  $V \setminus X_0$  the *central* respectively *peripheral* vertices of  $G$ .

(5) For a given natural number  $k$  we call a starlike graph  $G$   $k$ -starlike iff  $\beta_i \leq k$  for every  $i \in \{1, \dots, r\}$ .

(6) A starlike graph is called a *split graph* iff for every  $i \in \{1, \dots, r\}$   $\beta_i = 1$ .

Note that the classical definition for split graphs is a little different. For an overview see [9]. The class of split graphs is exactly the subclass of the class of chordal graphs whose complement is also chordal [5].

**Lemma 2.8.** *If  $G = (V, E)$  is a split graph then the pathwidth of  $G$  is  $\alpha_0$  or  $\alpha_0 - 1$ .*

**Proof.** In every path decomposition  $Y_i$  of  $G$  there is a set  $Y_s$  with  $X_0 \subseteq Y_s$ . So the pathwidth of  $G$  is at least  $\alpha_0 - 1$ . On the other hand it is at most  $\alpha_0$  since  $Y_i = X_i \cup X_0$  for  $i \in I$  is a path decomposition of  $G$ .  $\square$

### 3. Basic facts

We give some definitions and lemmas to characterize certain optimum path decompositions.

**Definition 3.1.** Let  $G$  be a starlike graph. We call a path decomposition  $(Y_j)$  of  $G$  *normalized* if every maximal clique  $X_i$  is contained in exactly one  $Y_{\varrho(i)}$ , i.e., there is a permutation  $\varrho : \{0, \dots, r\} \hookrightarrow \{0, \dots, r\}$  with  $X_i \subseteq Y_{\varrho(i)}$  and  $X_j \not\subseteq Y_{\varrho(i)}$  for  $j \notin \{0, i\}$ .

**Lemma 3.2.** *Every starlike graph  $G$  has an optimum path decomposition that is normalized.*

**Proof.** Let  $(Y_j)$  be an optimum path decomposition of  $G$ . Each  $X_i$ ,  $i \neq 0$ , has to appear in one of the  $Y_j$  since  $X_i$  is a maximal clique. If it appears in several of the  $Y_j$  just choose one of them and delete all nodes of  $X_i \setminus X_0$  in the others.

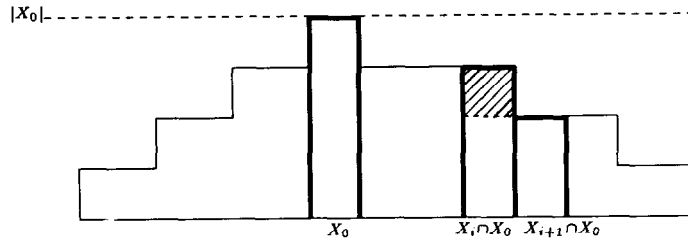
If several  $X_i$  are contained in the same  $Y_j$ , replace  $Y_j$  by as many copies of  $Y_j$  as  $X_i$  are contained in it and delete extra nodes as above.

If we have now  $X_0 = Y_j$  more than once, we choose an arbitrary one. If  $X_0 \neq Y_j$  for all  $i$  we put it before an arbitrary  $Y_j$  with  $X_0 \subset Y_j$ .

Since we only reduce the size of the  $Y_i$  by this procedure, the resulting path decomposition is still optimal and it has the desired properties.  $\square$

**Definition 3.3.** Let  $G = (V, E)$  be a starlike graph with fixed tree decomposition  $(T, \{X_0, \dots, X_r\})$ , an associated normalized path decomposition  $(Y_i)$  and an associated permutation  $\varrho$  s.t.  $X_i \subset Y_{\varrho(i)}$ .

Let  $\beta'_i = \beta_{\varrho^{-1}(i)}$  (cf. Definition 2.7). We call  $(Y_i)$  *sorted* if the numbers  $\beta'_i$  of the peripheral vertices in  $Y_i$  are decreasing respectively increasing on the left respec-

Fig. 1. A chart of the  $\alpha_i$ .

tively on the right of  $Y_{\varrho(0)}$ , i.e.,

$$\beta'_0 \geq \beta'_1 \geq \dots \geq \beta'_{s-1} \geq \beta'_s = \beta_0 = 0 \leq \beta'_{s+1} \leq \dots \leq \beta'_r.$$

**Lemma 3.4.** *Every starlike graph has an optimal path decomposition which is sorted.*

**Proof.** Let  $(Y_i)$  be a normalized optimal path decomposition of  $G$ . Observe that  $Y_i \cap X_0$  is increasing until  $X_0$  is reached and then decreasing. So if we draw a “chart” of the  $\alpha'_i$  we see a staircase pattern (see Fig. 1).

The order of the  $Y_i$  should be such that the  $Y_i \setminus X_0$  fit into that staircase in the right way (see Fig. 2).

Let  $Y_j$  and  $Y_{j+1}$  be such that  $X_0$  is on the left of them and suppose that  $\beta'_j > \beta'_{j+1}$ . Let  $\gamma_j = |Y_j \cap X_0|$ ,  $\gamma_{j+1} := |Y_{j+1} \cap X_0|$ . We have  $Y_{j+1} \cap X_0 \subseteq Y_j \cap X_0$ .

We have  $\gamma_j \geq \gamma_{j+1}$  since  $X_0$  is on the left.

We then may replace  $Y_j$  and  $Y_{j+1}$  by  $Y'_j := (Y_j \cap X_0) \cup Y_{j+1}$  and  $Y'_{j+1} := Y_j$  and obtain a feasible path decomposition. Since

$$|Y'_j| = \gamma_j + \beta'_{j+1} \leq \gamma_j + \beta'_j = |Y_j| = |Y'_{j+1}|$$

the resulting path decomposition is still optimal.

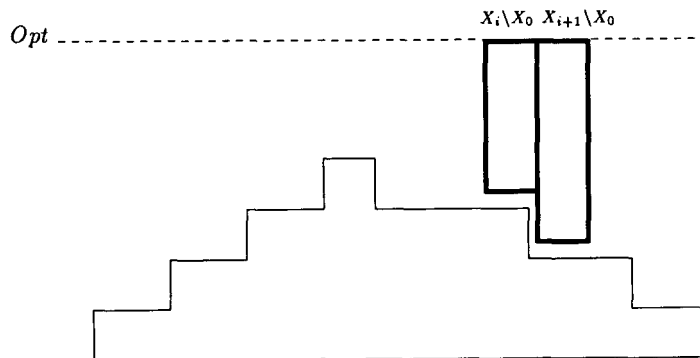


Fig. 2. Fitting into the staircase.

By repeating this argument we may replace our original path decomposition by a path decomposition in which each pair of neighbors fulfills the desired inequality and which is thus sorted.  $\square$

#### 4. NP-completeness

The three decision problems PWP, TWP and VSP have been shown to be NP-complete in various contexts:

PWP in [12] and [1] and TWP in [1].

VSP is equivalent to BALANCED COMPLETE BIPARTITE SUBGRAPH that was introduced by Garey and Johnson in [6] and shown to be NP-complete by Johnson in [10].

To see this equivalence observe that each balanced complete bipartite subgraph  $G' = (V_1, V_2, E')$  in the complement  $\bar{G}$  of the graph  $G$  induces a feasible solution  $X_1 = V \setminus V_2$ ,  $X_2 = V \setminus V_1$  of VSP.

Vice versa, if we have a feasible solution  $X_1, X_2$  of VSP the symmetric difference of  $X_1$  and  $X_2$  induces a bipartite subgraph of  $\bar{G}$ .

An easy calculation of the sizes of the solutions related in that way shows that we have optimality on one side iff we have it on the other side, too.

VSP is NP-hard even for a very strongly restricted class of graphs: it is even NP-hard for 3-regular graphs [19].

Our NP-completeness result is the following.

**Theorem 4.1.** *PWP is NP-complete for the class of chordal graphs.*

For the proof we give a reduction from VSP for arbitrary (finite) graphs to PWP for a subclass of the class of chordal graphs.

We will now define a mapping  $\varphi$  for the reduction. It assigns to an arbitrary graph  $G$  a starlike graph  $G'$ .  $V(G)$  forms the central clique of  $G'$  and every edge  $e$  in  $E(G)$  corresponds to a peripheral clique of  $G'$ . Such a peripheral clique consists of the two vertices of the edge and  $|V(G)|$  additional vertices. Each of these additional vertices occurs exactly in one such peripheral clique. For an example of this construction see Figs. 3 and 4.

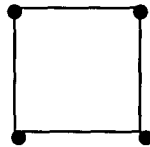
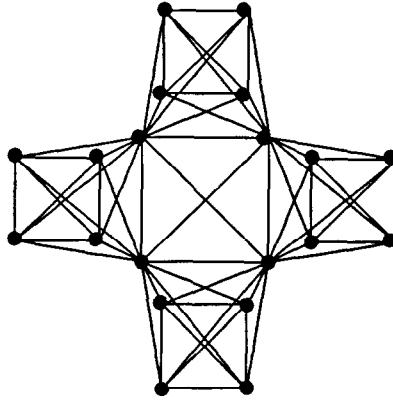


Fig. 3. The graph  $G$ .

Fig. 4. The graph  $G' := \varphi(G)$ .

Let  $e_1 = \{v_1, w_1\}, \dots, e_r = \{v_r, w_r\}$  be all edges of  $E$ , and let  $n = |V|$ . We define the chordal graph  $\varphi(G) = (V', E')$ , sets  $X_i \subseteq V'$  and the tree  $T = (I, E(T))$  as follows:

Define  $T$  to be the star  $K_{1,r}$  with center 0,  $I = \{0, \dots, r\}$  the set of nodes of  $T$  and  $I' = I \setminus \{0\}$  the set of leaves of  $T$ .

Define  $X_0 := V$ ,  $X_i = \{v_i, w_i\} \cup \{v_1^i, \dots, v_n^i\}$ ,  $i \in I$ , where

$$\{v_j^i \mid i = 1, \dots, n, j = 1, \dots, r\}$$

is an additional set of  $|V| \cdot |E|$  pairwise distinct additional vertices (i.e., vertices not in  $V$ ).

Define  $V' := \bigcup_{i \in I} X_i$  and let  $E'$  be the set of edges enforced by fulfilling (W2') —i.e., every  $X_i$  becomes a clique.

**Lemma 4.2.** (1) Every feasible solution  $(Z_1, Z_2)$  of VSP on a graph  $G = (V, E)$  with  $\max\{|Z_1|, |Z_2|\} = k$  induces a path decomposition  $Y_0, \dots, Y_r$  of  $G' = \varphi(G)$  with

$$\max_{i \in I} |Y_i| = k + |V|.$$

(2) Every normalized path decomposition  $Y_0, \dots, Y_r$  for  $G' = \varphi(G)$  with  $\max_{i=1, \dots, r+1} |Y_i| = l$ , induces a feasible solution  $Z_1, Z_2$  for VSP on  $G$  with

$$\max\{|Z_1|, |Z_2|\} = l - |X_0|.$$

**Proof.** To show (1), let  $Z_1, Z_2$ , be a feasible solution of VSP on  $G = (V, E)$ .

Let  $E_1 = \{e \in E \mid e \cap Z_1 \neq \emptyset\}$  and  $E_2 = E \setminus E_1$ .

W.l.o.g. the ordering of the edges of  $G$  is such that  $E_1 = \{e_1, \dots, e_s\}$  and  $E_2 = \{e_{s+1}, \dots, e_r\}$ . Let

$$Y_j = \begin{cases} X_{j+1} \cup Z_1 & \text{iff } e_{j+1} \in E_1, \\ X_j \cup Z_2 & \text{iff } e_j \in E_2, \\ X_0 & \text{iff } j = s. \end{cases}$$

This definition ensures that  $Y_0, \dots, Y_r$  is a path decomposition of  $G'$ . We have to show that the  $Y_i$  are not too big:

For  $j \leq s-1$  we have  $Y_j \cap X_0 = Z_1$ , so

$$|Y_j| = |Y_j \cap X_0| + |Y_j \setminus X_0| = |Z_1| + |V|.$$

Equivalently for  $j \geq s+1$  we have  $|Y_j| = |Z_2| + |V|$ . This proves (1).

For (2), let  $Y_1, \dots, Y_{r+1}$  be a normalized path decomposition for  $G'$ . Let  $s$  be such that  $X_0 \subseteq Y_s$  and  $Z_1 = Y_{s-1} \cap X_0$  and  $Z_2 = Y_{s+1} \cap X_0$ . It is clear that  $(Z_1, Z_2)$  is feasible for VSP and that has the appropriate size.  $\square$

Because every  $G' \in \text{Im}(\varphi)$  is starlike and  $\varphi$  can be calculated in polynomial time we get

**Corollary 4.3.** *PWP is NP-complete for the class of starlike graphs.*

That completes our proof of Theorem 4.1.

## 5. A generalized partition problem

We will solve PWP for some classes of starlike graphs by showing equivalence to the following problem on natural numbers.

**Definition 5.1.** The *generalized partition problem*—GPP for short—is the following:

*Instance:* Nonnegative numbers  $r, \alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_r$ .

*Problem:* Find a permutation  $\pi: \{0, \dots, r\} \rightarrow \{0, \dots, r\}$  s.t. the maximum of the values

$$\beta_i + \sum_{\substack{j \in \{1, \dots, r\} \\ \pi(j) \leq \pi(i)}} \alpha_j, \quad \text{for } \pi(i) < \pi(0),$$

$$\beta_i + \sum_{\substack{j \in \{1, \dots, r\} \\ \pi(j) \geq \pi(i)}} \alpha_j, \quad \text{for } \pi(i) > \pi(0)$$

is minimum.

GPP is a generalization of the classical partition problem (PP) which may be seen as GPP with  $\beta_1 = \dots = \beta_r = 0$ . It was shown by Karp to be NP-hard [11]. A pseudopolynomial algorithm for PP was given by Garey and Johnson in [6]. We will give a pseudopolynomial dynamic programming algorithm for GPP which is based on similar ideas as the algorithm for PP.

**Remark 5.2.** Every optimal solution  $\pi$  of GPP can be modified to an optimal solution s.t.

$$\beta_{\pi(i)-1} \geq \beta_{\pi(i)}, \quad \text{if } \pi(i) < \pi(0)$$

and

$$\beta_{\pi(i)} \leq \beta_{\pi(i)+1}, \quad \text{if } \pi(0) < \pi(i).$$

For a proof proceed analogously to Lemma 3.4.

**Definition 5.3.** (1) A *sorted generalized partition problem* (SGPP) is a GPP whose input is sorted s.t.

$$\beta_i \geq \beta_j, \quad \text{for } i \leq j.$$

(2) For a SGPP given by  $(\alpha_1, \dots, \alpha_r), (\beta_1, \dots, \beta_r)$  let

- $s_k := \sum_{i=1}^k \alpha_i$ .
- $Opt_k$  denote the value of an optimal solution when the problem is restricted to  $(\alpha_1, \dots, \alpha_k, s_r - s_k), (\beta_1, \dots, \beta_k, 0)$ .
- $Opt_k(s)$ , for  $s \leq s_r$ , be defined as  $Opt_k$  but with the additional assumption that

$$\sum_{\pi(i) < \pi(0)} \alpha(i) = s.$$

Set  $Opt_k(s) = \infty$  if such a solution does not exist.

- $Opt_k^c(s)$ ,  $s \leq s_r$ ,  $c \in \{+, -\}$ , be defined as  $Opt_k(s)$  but with the additional assumption that  $\pi(k) < \pi(0)$  for  $c = -$  and  $\pi(k) > \pi(0)$  for  $c = +$ .

**Lemma 5.4.** (1)  $Opt_k = \min_s Opt_k(s)$ .

(2)  $Opt_k(s) = \min\{Opt_k^-(s), Opt_k^+(s)\}$ .

(3)  $Opt_k^-(s) = \max\{Opt_{k-1}(s - \alpha_k), s + \beta_k\}$ .

(4)  $Opt_k^+(s) = \max\{Opt_{k-1}(s), s_k - s + \beta_k\}$ .

**Proof.** (1) and (2) are clear. We now show (3).

If we have an optimal solution  $\pi$  for  $Opt_{k-1}(s - \alpha_k)$  we can easily extend it to a solution for  $Opt_k^-(s)$  by inserting  $\pi(k)$  just before  $\pi(0)$  in  $\pi$ . This gives “ $\leq$ ” in (3).

To show “ $\geq$ ” observe that we always have  $Opt_k^-(s) \geq s + \beta_k$ .

Now let  $\pi$  be an optimal solution for  $Opt_k^-(s)$ , and let  $\pi'$  be the resulting permutation after omitting  $\pi(k)$ .  $\pi'$  is a solution for  $Opt_{k-1}(s - \alpha_k)$ . The value of  $\pi'$  is smaller than or equal to that of  $\pi$ . So “ $\geq$ ” holds for the optimum values.

Finally, (4) follows by symmetry.  $\square$

The recursion formulas of Lemma 5.4 lead directly to the following algorithm.

**Algorithm 5.5 (SGPP).**

*Input:* Nonnegative number  $r$ , sequences  $\alpha_1, \dots, \alpha_r$  and  $\beta_1, \dots, \beta_r$  of nonnegative numbers where the  $\beta_i$  are decreasing.

*Output:* Nonnegative number  $Opt$ , the value of an optimal solution of the corresponding SGPP.

1.  $Opt_0(0) := 0; s_0 := 0;$
2. **for**  $k := 1$  **to**  $r$  **do**  $s_k := s_{k-1} + \alpha_k;$
3. **for**  $k := 1$  **to**  $r$  **do begin**
4.     **for**  $s := 0$  **to**  $s_k$  **do begin**
5.          $Opt_k^-(s) := \max\{Opt_{k-1}(s - \alpha_k), s + \beta_k\};$
6.          $Opt_k^+(s) := \max\{Opt_{k-1}(s), s_k - s + \beta_k\};$
7.          $Opt_k(s) := \min\{Opt_k^-(s), Opt_k^+(s), \infty\}$
8.     **end;**
9. **end;**
10.  $Opt := \min_{0 \leq s \leq s_r} Opt_r(s);$

The correctness of this algorithm for SGPP and its complexity are straightforward and summarized in the following theorem.

**Theorem 5.6.** *SGPP is solvable in  $O(r \cdot s_r)$  pseudopolynomial time and space.*

Observe that the values  $\beta_1, \dots, \beta_r$  do not occur in the complexity of the algorithm. To solve GPP we have to sort these values according to their size.

This sorting can be done in  $O(r + s_r)$  time. Thus we have

**Corollary 5.7.** *GPP is solvable in  $O(r \cdot s_r)$  pseudopolynomial time and space.*

With this result we obtain the following

**Theorem 5.8.** *The pathwidth of a primitive starlike graph  $G$  can be calculated in  $O(r \cdot \alpha_0) = O(|V(G)|^2)$  time.*

**Proof.** Let  $(\alpha_1, \dots, \alpha_r), (\beta_1, \dots, \beta_r)$  be as in Definition 2.7(3) and let  $\pi$  be an optimal solution of the corresponding GPP. Define

$$Y_i = \begin{cases} (X_{\pi^{-1}(i)} \setminus X_0) \cup \left( \bigcup_{\pi(j) \leq i} X_j \cap X_0 \right), & \text{for } 0 \leq i < \pi(0), \\ (X_{\pi^{-1}(i)} \setminus X_0) \cup \left( \bigcup_{\pi(j) \geq i} X_j \cap X_0 \right), & \text{for } \pi(0) < i \leq r, \\ X_0, & \text{for } i = \pi(0). \end{cases}$$

$Y_0, \dots, Y_r$  is a path decomposition of  $G$ . It has to be optimum since a better path decomposition would lead to a better solution for SGPP, a contradiction.

For the complexity observe that  $r + 1$  (the number of maximal cliques of  $G$ ) and  $s_r$  are smaller than  $|V|$ .  $\square$

Note that Theorem 5.8 and its proof is easily extended to the class of graphs for which Definition 2.7(2a) is replaced by

$$X_i \cap X_j \neq \emptyset \Rightarrow X_i \cap X_0 = X_j \cap X_0.$$

## 6. An exact algorithm for starlike graphs

The algorithm we designed for GPP can be generalized to arbitrary starlike graphs. In general it needs exponential time—as one would expect for an NP-hard problem.

We define optimal values for path decompositions with certain restrictions analogously to Definition 5.3.

**Definition 6.1.** Let  $G$  be a  $k$ -starlike graph given by its tree decomposition  $X_0, X_1, \dots, X_r$ . We assume

$$\beta_i \geq \beta_{i+1} \quad \forall i, i > 0.$$

- Let  $Opt_m$  be the pathwidth of the problem restricted to  $X_0, \dots, X_m$ .
- For the problem restricted to  $X_0, \dots, X_m$  and sets  $S^-, S^+ \subseteq X_0$  let  $Opt_m(S^-, S^+)$  be the minimum size of a path decomposition  $(Y_i)$ ,  $X_{\pi(i)} \subseteq Y_i$ , with the additional assumption that

$$\bigcup_{\pi(i) < \pi(0)} (Y_i \cap X_0) = S^- \quad \text{and} \quad \bigcup_{\pi(i) > \pi(0)} (Y_i \cap X_0) = S^+.$$

Set  $Opt_m(S^-, S^+) = \infty$  if such a path decomposition does not exist.

- We say that  $X_i$  is on the *left* respectively *right* in that decomposition if  $\pi(i) < \pi(0)$  respectively  $\pi(i) > \pi(0)$ .
- Let  $Opt_m^c(S^-, S^+)$  with  $c \in \{-, +\}$  be as  $Opt_m(S^-, S^+)$  but with the additional assumption that  $X_m$  is on the left of  $X_0$  if  $c = -$  and on the right if  $c = +$ .

Observe that  $S^-$  and  $S^+$  replace the natural number  $s$  in Definition 5.3. We call  $S^-$  respectively  $S^+$  the *left-* respectively *rightconstraint* of the restricted problem. The reason for the use of sets is that now the  $X_i$  may overlap and we need more information than the cardinality of the sets  $X_i \setminus X_0$  and  $X_i \cap X_0$ .

We get a lemma analogously to Lemma 5.4.

**Lemma 6.2.** (1)  $Opt_m = \min_{(S^-, S^+)} Opt_m(S^-, S^+)$ .

(2)  $Opt_m(S^-, S^+) = \min\{Opt_m^-(S^-, S^+), Opt_m^+(S^-, S^+)\}$ .

(3) With  $Min_m^-(S^-, S^+) = \min\{Opt_{m-1}(T^-, S^+) \mid S^- = T^- \cup (X_m \cap X_0)\}$ , we obtain

$$Opt_m^-(S^-, S^+) = \max\{|S^- \cup X_m|, Min_m^-(S^-, S^+)\}.$$

(4) With  $Min_m^+(S^-, S^+) = \min\{Opt_{m-1}(S^-, T^+) \mid S^+ = T^+ \cup (X_m \cap X_0)\}$ , we obtain

$$Opt_m^+(S^-, S^+) = \max\{|S^+ \cup X_m|, Min_m^+(S^-, S^+)\}.$$

**Proof.** (1) and (2) are again trivial.

We show (3) “ $\leq$ ”.

Let  $T^-$  be such that  $S^- = T^- \cup (X_m \cap X_0)$  and suppose we have a path decomposition  $(Y_i)$  which attains the optimum value  $Opt_{m-1}(T^-, S^+)$ .

Take  $Y_m = X_m \cup S^-$  and put it directly before  $X_0$ . The resulting path decomposition for  $Y'_0, \dots, Y'_m$  fulfills all desired properties and has size

$$\max\{|S^- \cup X_m|, Opt_{m-1}(T^-, S^+)\}.$$

This shows “ $\leq$ ”.

To show “ $\geq$ ” choose an optimum path decomposition for the problem restricted to  $X_0, \dots, X_m$  constraints  $S^-, S^+$  and with  $X_m$  on the left of  $X_0$ . Omitting the corresponding  $Y_m$  leads to a path decomposition for  $X_0, \dots, X_{m-1}$  for some constraints  $T^-$  and  $S^+$ .

So we have that  $Opt_m^-(S^-, S^+)$  is not smaller than the considered minimum. Since it is obviously greater than  $|S^- \cup X_m|$  we have “ $\geq$ ”, too.

(4) follows again by symmetry.  $\square$

### Algorithm 6.3.

*Input:* Nonnegative number  $r$ , maximal cliques of a starlike graph  $X_0, \dots, X_m$  s.t. the sequence  $|X_1 \setminus X_0|, \dots, |X_m \setminus X_0|$  is decreasing.

*Output:* Nonnegative number  $Opt$ , the value of an optimal solution of the corresponding PWP.

1. Preprocessing;
2. Initialization;
3. **for**  $m := 1$  **to**  $r$  **do begin**
4.     **for all**  $T^-, T^+ \subseteq X_0$  **begin**
5.          $S^- = T^- \cup (X_m \cap X_0); S^+ = T^+ \cup (X_m \cap X_0);$
6.          $Min_m^-(S^-, T^+) = \min(Min_m^-(S^-, T^+), Opt_{m-1}(T^-, T^+));$
7.          $Min_m^+(T^-, S^+) = \min(Min_m^+(T^-, S^+), Opt_{m-1}(T^-, T^+))$
8.     **end;**
9.     **for all**  $S^-, S^+ \subseteq X_0$  calculate  $Opt_m^-(S^-, S^+)$ ,  $Opt_m^+(S^-, S^+)$  and  $Opt_m(S^-, S^+);$
10.    **end;**
11.  $Opt := \min_{(S^-, S^+)} Opt_r(S^-, S^+);$

Initialization here means to set

$$Opt_0(S, T) = \begin{cases} \alpha_0 & \text{iff } S = T = \emptyset, \\ \infty & \text{otherwise,} \end{cases}$$

and all  $Opt_m^\pm(S^-, S^+)$  and  $Min_m^\pm(S^-, S^+)$  for  $m \geq 0$  to  $\infty$ .

The preprocessing that must ensure that we can access  $S = T \cup (X_m \cap X_0)$  as needed in Step 5 can be done in constant time. First we construct all the sets  $T$  via

an  $\alpha_0$ -dimensional array and enumerate them. Then we may calculate all  $S = T \cup (X_m \cap X_0)$  and store the appropriate number into an array  $Union[Number(T), m]$ .

This initialization and preprocessing can be done in  $O((2^{\alpha_0})^2 \cdot r)$  time.

We now have:

**Theorem 6.4.** *The pathwidth of a starlike graph may be calculated in  $O(((2^{\alpha_0})^2 \cdot r) + |V|)$  time.*

**Proof.** To apply Algorithm 6.3 we have to sort the  $\beta_i$ . With bucketsort this can be done in  $O(|V| + r)$  time, since the  $\beta_i$  are bounded by  $|V|$ .

The correctness of Algorithm 6.3 is a straightforward application of Lemma 6.2. Just observe that the calculation of  $Min_m^\pm(S^-, S^+)$  in loop 4 leads to the correct value.  $\square$

Observe that this leads also to polynomial algorithms for classes of starlike graphs with  $\alpha_0$  bounded by a constant or by a polynomial in  $\log |V|$ .

## 7. $k$ -starlike graphs

We will now apply Algorithm 6.3 to obtain the following result on  $k$ -starlike graphs.

**Theorem 7.1.** *For a  $k$ -starlike graph  $G$ , PWP can be solved in  $O(\alpha_0^{2k} \cdot r) = O(|V(G)|^{2k+1})$  time and space.*

For the proof we need

**Definition 7.2.** For a  $k$ -starlike graph  $G$  and  $0 \leq l \leq k$  we denote by  $G_l$  the subgraph of  $G$  induced by the cliques with small peripheral size, i.e., by

$$V(G_l) = \bigcup_{\beta_i \leq l} X_i.$$

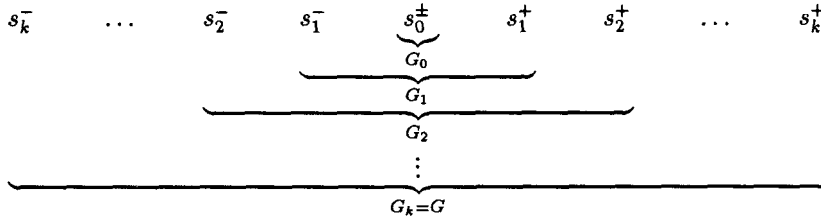
Observe that  $G_0$  is  $K_{\alpha_0}$ ,  $G_k = G$  and that all  $G_l$  are  $l$ -starlike.

With Lemma 3.4 we see also that  $G$  has an optimum path decomposition that looks like Fig. 5.

This means that for a given sorted optimal path decomposition, there is for each  $l$  an interval  $[s_l^-, s_l^+]$  s.t.  $Y_{s_l^-}, \dots, Y_{s_l^+}$  is a path decomposition for  $G_l$ .

We will extend such an optimal path decomposition such that all  $Y_i \cap X_0$  will not be too small.

**Lemma 7.3.** *Let  $G$  be a  $k$ -starlike graph. Every sorted path decomposition  $(Y_i)$  of*

Fig. 5. A nested path decomposition for the  $G_i$ .

$G$  with size  $\alpha_0 + l$  can be modified to a path decomposition  $(Y'_i)$  of the same size with

$$Y_i \subseteq Y'_i$$

and

$$|Y'_i \cap X_0| \geq \alpha_0 + l - k \quad \text{for all } i.$$

**Proof.** The statement is trivial for  $l = k$  and  $k = 0$ . We proceed for induction on  $k$ . The given path decomposition  $(Y_i)$  induces a sorted path decomposition  $(Z_i)$  for  $G_{k-1}$  of size  $\alpha_0 + l_0$  with  $l_0 \leq l$ . If  $l_0 < l$  we may extend  $(Z_i)$  on each side of  $X_0$  by an arbitrary subset of  $X_0$  such that it has size  $\alpha_0 + l$ . So we may assume that  $l_0 = l$  and that  $(Z_i)$  fulfills the inductive hypothesis for  $k - 1$ . Let  $Z^-$  and  $Z^+$  be the leftmost respectively rightmost set in  $(Z_i)$ .

There are nodes  $v^- \in Z^- \cap X_0$ ,  $v^+ \in Z^+ \cap X_0$  s.t. for  $i$  with  $|Y_i \setminus X_0| = k$ ,

$$v^- \notin Y_i, \quad \text{if } Y_i \text{ is on the left of } X_0,$$

$$v^+ \notin Y_i, \quad \text{if } Y_i \text{ is on the right of } X_0.$$

Set  $Y^- = (Z^- \cap X_0) \setminus \{v^-\}$ ,  $Y^+ = (Z^+ \cap X_0) \setminus \{v^+\}$  and set

$$Y'_i = \begin{cases} Y_i \cup Y^-, & \text{if } |Y_i \setminus X_0| = k \text{ and it is on the left,} \\ Y_i \cup Y^+, & \text{if } |Y_i \setminus X_0| = k \text{ is on the right,} \\ Z_i, & \text{otherwise.} \end{cases}$$

$Y'_i$  is a path decomposition of  $G$  with the desired properties.  $\square$

**Proof of Theorem 7.1.** Based on this lemma observe that the sets which can be left-respectively rightconstraints  $S^-$  respectively  $S^+$  have cardinality  $\geq \alpha_0 - k$ . So there are at most

$$\sum_{l=1}^k \binom{\alpha_0}{l} \leq k \cdot \binom{\alpha_0}{k} = O(\alpha_0^k)$$

such sets.

Algorithm 6.3 can easily be adapted to use only those sets. So the complexity is  $O(r \cdot \alpha_0^{2k})$  which can be estimated by  $O(|V| \cdot |V|^{2k})$ .  $\square$

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